Practical No. 8

Apply Defect Management, Test Case Management and Execution Report Generation using Bugzilla and TestLink Workflows

Aim

To apply defect management, test case management, and test execution reporting for the Login module of an Online Shopping System, using TestLink for test case design and execution, and Bugzilla for logging and tracking defects.

Tools Used

- TestLink 1.9.20 (BETA) – Test case management and test execution tool
- Bugzilla – Defect/bug tracking tool

Theory

Test case management involves designing, organizing, and executing test cases in a structured manner so that the coverage and results of testing can be tracked over time. TestLink allows testers to create test plans, add test cases under a project, assign builds, and record execution results (Pass/Fail/Blocked) against each test case.

Defect management is the process of identifying, reporting, tracking, and resolving defects found during testing. Bugzilla is used to log a defect with details such as product, component, severity, priority, steps to reproduce, expected result, and actual result, so that developers can reproduce and fix the issue, and testers can verify the fix.

In this practical, test cases for the Login module were created and executed in TestLink. During execution, the 'Invalid Login' test case failed because the application did not display an error message for invalid credentials. This defect was then logged in Bugzilla as Bug #1, linking the test management and defect management workflows together.

Environment / Project Details

Test Project	Online Shopping System
Test Plan	Login Module Test Plan
Build	Build 1.0
Component (Bugzilla)	Login
Tester	Saloni Ashish Tanna
TestLink URL	http://localhost/testlink/index.php
Bugzilla URL	http://localhost/bugzilla/

Part A: Test Case Management & Execution using TestLink

Step 1: Login to TestLink

The tester logs into TestLink using valid administrator credentials to access the project workspace.



Fig. A1 – TestLink Login Page

Step 2: Project Dashboard

After logging in, the TestLink dashboard shows the recent activity for the 'Online Shopping System' project, including creation of the test plan and test cases, and execution history.

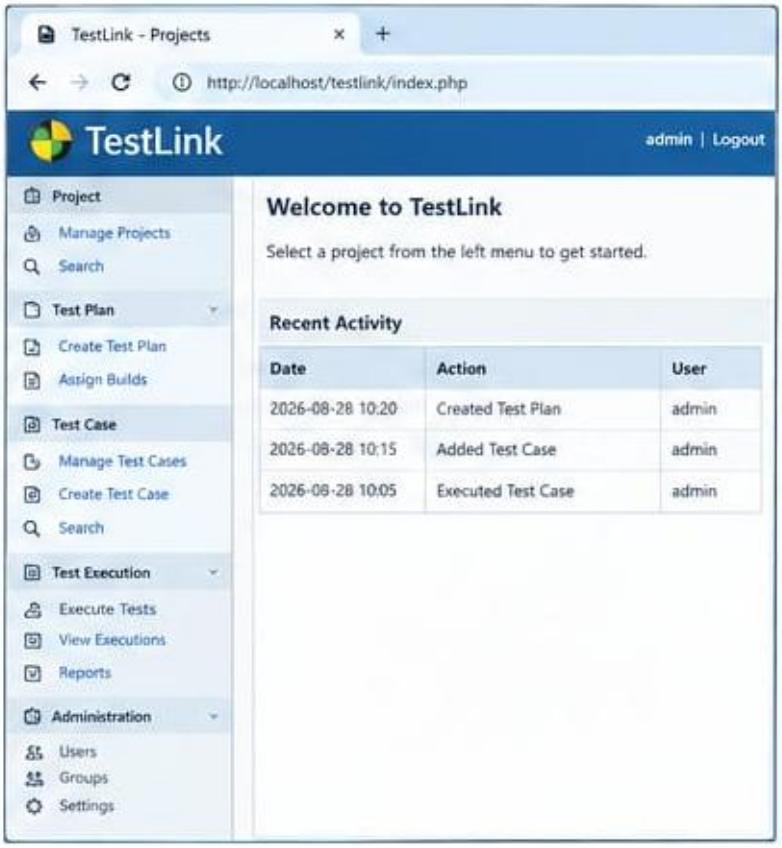


Fig. A2 – TestLink Project Dashboard

Step 3: Test Case List

Eight test cases (TC-001 to TC-008) were designed for the Login and related modules of the Online Shopping System and added under 'Manage Test Cases' in TestLink, covering login, password recovery, logout, cart, and order management scenarios.

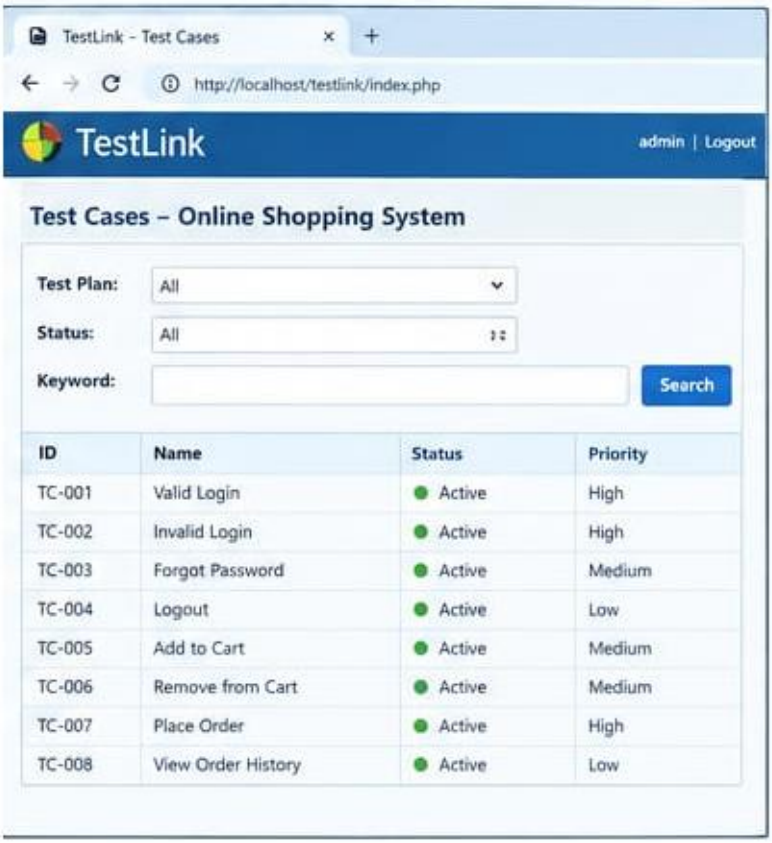


Fig. A3 – Test Case List for Online Shopping System

Step 4: Execute Test Case

Each test case is executed against Build 1.0 under the Login Module Test Plan. The screenshot below shows the execution of TC-001 (Valid Login), which was marked as 'Passed' with the note 'Login works as expected.'

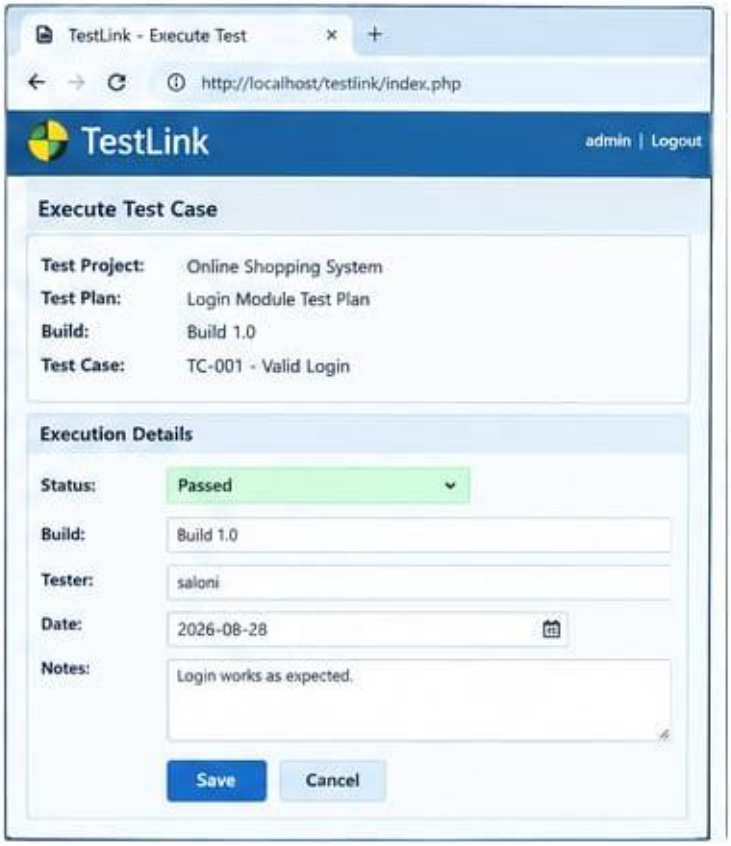


Fig. A4 – Executing TC-001 (Valid Login) in TestLink

Test Execution Report (Summary)

The table below summarizes the expected and actual results recorded for all eight test cases executed under the Login Module Test Plan.

Test Case ID	Test Case Name	Expected Result	Actual Result	Status
TC-001	Valid Login	User logged in and redirected to dashboard	Logged in successfully; redirected to home page	Pass
TC-002	Invalid Login	Proper error message shown for invalid credentials	Login failed but no error message was displayed	Fail
TC-003	Forgot Password	Reset instructions sent to registered email	Reset instructions sent successfully	Pass
TC-004	Logout	User logged out, redirected to login page	Logged out and redirected correctly	Pass
TC-005	Add to Cart	Product added to cart	Product added; cart count updated	Pass
TC-006	Remove from Cart	Product removed, totals updated	Product removed; totals updated correctly	Pass
TC-007	Place Order	Order placed, confirmation shown	Order placed with confirmation message	Pass
TC-008	View Order History	Past orders displayed with correct details	Order history displayed correctly	Pass

Total Test Cases	8
Passed	7
Failed	1
Pass %	87.5%

Part B: Defect Management using Bugzilla

During execution of TC-002 (Invalid Login), the application failed to display an error message for invalid credentials. This defect was reported in Bugzilla as described below.

Step 1: Login to Bugzilla

The tester (saloni) logs into Bugzilla to report and manage defects for the 'Online Shopping System' product.



Fig. B1 – Bugzilla Login Page

Step 2: Enter New Bug

A new bug is entered against Product 'Online Shopping System', Component 'Login', with the summary 'Login page throws error on invalid credentials', including steps to reproduce, expected result, and actual result.



Fig. B2 – Entering a New Bug (Defect Report)

Step 3: View Bug

Once submitted, the bug is assigned Bug ID #1 and can be viewed with full details, current status (UNCONFIRMED), and reporter information.



Fig. B3 – Viewing Bug #1 Details

Step 4: Browse Bugs

The 'Browse Bugs' page lists all logged bugs for the Login component of the Online Shopping System, allowing filtering by status, product, component, and summary.



Fig. B4 – Browse Bugs List

Bug Report

Field	Description
Bug ID	1
Product	Online Shopping System
Component	Login
Summary	Login page throws error on invalid credentials
Severity	Normal
Priority	P2
Steps to Reproduce	1. Go to the login page. 2. Enter invalid username and password. 3. Click on the login button.
Expected Result	A proper error message should be shown
Actual Result	Error message is not displayed
Status	UNCONFIRMED
Reporter	saloni

Conclusion

Test cases for the Login module of the Online Shopping System were successfully designed, organized, and executed using TestLink, and the results were recorded against Build 1.0 of the Login Module Test Plan. Out of 8 test cases executed, 7 passed and 1 failed, giving a pass rate of 87.5%. The failure in TC-002 (Invalid Login), where no error message was shown for invalid credentials, was logged as Bug #1 in Bugzilla with complete steps to reproduce, expected result, and actual result. This practical demonstrates how test case management (TestLink) and defect management (Bugzilla) tools work together in a typical software testing workflow: designing and executing test cases, identifying failures, and tracking the resulting defects through to resolution.